

The validation of a Polish version of Co-Presence and Social Presence Scale

Radosław Sterna*
Emotion and Perception Lab
Institute of Psychology
Jagiellonian University in Kraków
Doctoral School in the Social
Sciences
Jagiellonian University in Kraków

Michał Kuniecki §
Emotion and Perception Lab
Institute of Psychology
Jagiellonian University in Kraków

Natalia Lipp†
Sano Center
for Computational Personalised
Medicine
Kraków

Agnieszka Strojny‡
Institute of Applied Psychology
Jagiellonian University in Kraków

Paweł Strojny¶
Institute of Applied Psychology
Jagiellonian University in Kraków

ABSTRACT

This study validates the Polish translation of the Co-presence and Social Presence scale, confirming strong reliability and validity through robust internal consistency and favorable confirmatory factor analysis fit indices. Convergent validity results align with expectations, although correlation magnitudes are lower than anticipated. Examining discriminant validity, an unexpected weak positive correlation with eeriness challenges initial expectations of negative relationship. Furthermore, moderate to low correlations between co-presence and presence emphasize their distinct yet related nature

Index Terms: Virtual Reality (VR)—psychology—co-presence—virtual character;

1 INTRODUCTION

Our focus here is on co-presence, a subjective experience describing the sensation of being in the company of other social actors within the virtual environment [1]. Despite over 40 instruments developed to measure co-presence, these tools often lack rigorous testing for reliability and validity [9]. This scarcity of validated measures leads to reduced comparability across studies, hindering research progress. In response to this challenge, we aimed to validate the Polish version of Co-Presence and Social Presence Scale (C-PS [6]), a psychometrically tested scale originally designed for immersive virtual social environments. The C-PS scale comprises 15 items rated on a 5-point Likert scale and 4 subscales: Presenter's Reaction to Virtual Agents, Perceived Virtual Agents' Reaction, Impression of Interaction Possibilities, and (Co-)Presence of other people. Its use extends beyond its original context of public speaking scenario, providing a valuable resource for both researchers and developers. By retesting its quality and making it available for broader applications, we aim to contribute to the improvement of measurement standards in the field, fostering more reliable and comparable findings. In choosing the C-PS scale, we considered its VR-centric focus, roots in influential works, and alignment with multifaceted understandings of social presence [1].

*e-mail: radoslawsterna@gmail.com

†e-mail: n.lipp@sanoscience.org

‡e-mail: agnieszka.strojny@uj.edu.pl

§e-mail: michal.kuniecki@uj.edu.pl

¶e-mail: pawel.strojny@uj.edu.pl

2 TRANSLATION PROCESS

The translation process description was uploaded to: OSF link

3 PARTICIPANTS AND STUDIES DESCRIPTION

The participants and studies description was uploaded to: OSF link

4 MEASURES

Following questionnaires were used during this validation:

Set of questions related to appeal, eeriness and familiarity of a virtual character [11] consists of 3 items, originally rated on a 7-point Likert scale, but for the purpose of the study 2 it was modified to a 5-point Likert scale from -2 to 2 to follow the pattern of response of other questionnaires. Polish adaptation of the Igroup Presence Questionnaire (IPQ [8]) Polish translation by [10]) consists of 13 items and was used to measure the sense of presence (in three dimensions: spatial presence, involvement, and realness). The Polish version of the scale was shown to have a satisfying internal consistency coefficients (Cronbach's alpha for the three dimensions > .80).

Polish adaptation of the Realism scale ([5] Polish translation by [4]) consists of 14 items and was used to assess the realism perception in 4 dimensions: scene realism, audience behavior, audience appearance, sound realism. The Polish version of the scale has been shown to have satisfactory reliability (for combined scale Cronbach's alpha > .80)

5 ANALYSES

The analysis encompassed three primary steps. Firstly, to demonstrate the psychometric properties of the C-PS scale, we assessed its internal consistency and structure. Internal consistency was examined using Cronbach's alpha, applying Nunally's criterion of 0.70 as an indicator of good reliability for each of the four subscales.

Secondly, a Confirmatory Factor Analysis (CFA) was performed to verify if the Polish version of the scale maintained the same structure as the original. We employed a bifactor model, testing for co-presence's general sense and four orthogonal group factors: Presenter's Reaction to Virtual Agents, Perceived Virtual Agents' Reaction, Impression of Interaction Possibilities, and (Co-)Presence of other people. Model evaluation relied on TLI, CFI, SRMR, and RMSEA indices, adhering to established criteria (TLI > 0.95, CFI > 0.95, RMSEA < 0.08, SRMR < 0.08).

Thirdly, to establish convergent and discriminant validity, correlational analyses were conducted between C-PS general scores and other constructs. Convergent validity was tested through correlations with the Realism questionnaire's Audience Behavior, Appearance, and Scene Realism subscales. For discriminant validity, correlations were examined with the Immersive Presence Questionnaire (IPQ) and its subscales, expecting low to moderate correlations. Eeriness

was hypothesized to weakly correlate negatively with co-presence, given its association with unpleasant sensations in the virtual space from which the participants would try to cut off. The Spearman rank correlation analysis was used for one-item scales (Appeal, Familiarity, and Eeriness), and parametric analyses (Pearson's correlation) were applied for other correlations.

6 RESULTS

The overall Cronbach's alpha coefficient was found to be satisfactory (alpha = 0.885). Additionally, we calculated alpha coefficients for each of the four subscales: Presenter's Reaction to Virtual Agents (alpha = 0.761), Perceived Virtual Agents' Reaction (alpha = 0.789), Impression of Interaction Possibilities (alpha = 0.813), and (Co-)Presence of other people (alpha = 0.834) subscale. These obtained values also met the satisfactory threshold.

The results of the Confirmatory Factor Analysis show that tested model has a reasonably good fit: TLI = 0.929 (good but slightly below the expected threshold of 0.95), CFI = 0.956 (good, meeting the expected criterion of above 0.95), RMSEA = 0.071 (good, below the expected threshold of 0.08), and SRMR = 0.040 (good, below the expected threshold of 0.08). While two absolute indices (RMSEA and SRMR) and one of the incremental indices (CFI) suggest an acceptable fit, only the TLI falls slightly short of the predefined criteria. Please see Figure 1 in supplementary files for the model with estimated loadings.

For convergent validity, we observed low to moderate correlations between co-presence general score and measures of scene realism, behavior realism, and appearance realism, as well as moderate correlations with appeal and familiarity. For divergent validity C-PS general score correlated weakly or moderately with IPQ general score and its subscales. Unexpectedly, eeriness correlated weakly positively with C-PS general score. For detailed results please see supplementary materials.

7 DISCUSSION

This study validates the Polish translation of the Co-presence and Social Presence scale, confirming satisfactory reliability and validity. The reliability analysis, including Cronbach's alpha coefficients, demonstrated robust internal consistency for both the overall scale and individual subscales, affirming the measure's reliability in assessing co-presence.

Confirmatory factor analysis supported the structural integrity of the C-PS, with favorable fit indices such as CFI, RMSEA, and SRMR indicating an acceptable model fit. Although TLI fell slightly below predefined criteria, it aligns with other sources' perspectives on good fit (e.g. [3]).

In terms of convergent validity, correlations between the C-PS general score and various measures, while generally aligned with expectations, exhibited lower magnitudes than anticipated. This observation is common in psychological research as supported by meta-analytical data indicating that a significant portion of studies report correlation coefficients below 0.33 [2].

In examining discriminant validity, our study revealed a weak positive correlation with eeriness, contrary to our initial expectation of a negative relationship. The unexpected outcome suggests that the virtual characters in the study 2 may not have effectively triggered the Uncanny Valley effect, as reflected in the median eeriness rating of -1 in Study 2. The uniform human-like appearance of our virtual characters, unlike the diverse range used by [7], may contribute to this outcome.

As anticipated, our study revealed moderate to low correlations between co-presence and presence, evidenced by C-PS general score and IPQ scores. This expected correlation underscores that, while related, co-presence and presence represent distinct facets of the virtual reality experience, each with its unique dimensions of significance.

The validation of the Polish version of the C-PS scale can support local gaming and VR industry which is experiencing exponential growth, as shown by e.g. the global success of CD Projekt RED. A validated measure equips researchers and professionals in the gaming and VR industry with a robust tool to assess and enhance the immersive quality of virtual environments.

8 SUPPLEMENTARY FILES

All the supplementary files including the scale itself were uploaded to: OSF link

9 ACKNOWLEDGMENTS

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